

UserStory A1

1. Project Description

The goal of this task was to manufacture a **PVC display mount and a PVC base plate** for the Portable Indoor Feedback system.

Both parts needed to fit into a wooden frame and provide mounting points for:

- The touch display
- The Raspberry Pi
- Internal electronics

2. Materials & Tools

- PVC plates (for display holder & base plate)
- Drill bits:
 - 2.5 mm (for Raspberry Pi mounting & display mounting)
 - 2.3 mm (for M3 threads in the base plate)
 - 3 mm (for M3 screws through the display bracket)
 - 4 mm with countersink (for later wood screw mounting)
- Countersinking tool
- M2.5 screws, M3 countersunk screws, spacer sleeves
- Tools for deburring
- Workshop safety equipment

3. Manufacturing Process

3.1 Preparing the PVC Base Plate

1. **Drilling holes for Raspberry Pi & circuit board**
 - Holes drilled with **2.5 mm diameter** for M2.5 screws.
2. **Drilling threaded holes for attaching the PVC display holder**
 - Threads required: **M3**
 - Drill diameter: **2.3 mm** (warning noted in instructions)
3. **Drilling the mounting hole for later attachment to the wooden housing**
 - Diameter: **4 mm**, with countersink

3.2 Preparing the PVC Display Holder

1. **Drilling holes for the M3 through-screws**
 - Diameter: **3 mm**
 - Used to connect display bracket to the base plate
2. **Creating the ribbon-cable opening**
 - Opening cut according to provided specifications

4. Finishing Steps

1. **Deburring**
 - All holes and edges were **professionally deburred** to remove sharp edges.
2. **Countersinking**
 - Countersinks for all M3 countersunk screws were produced according to specifications.
3. **Marking the parts**
 - All manufactured parts were labeled with the personal IAM identification code.

5. Assembly

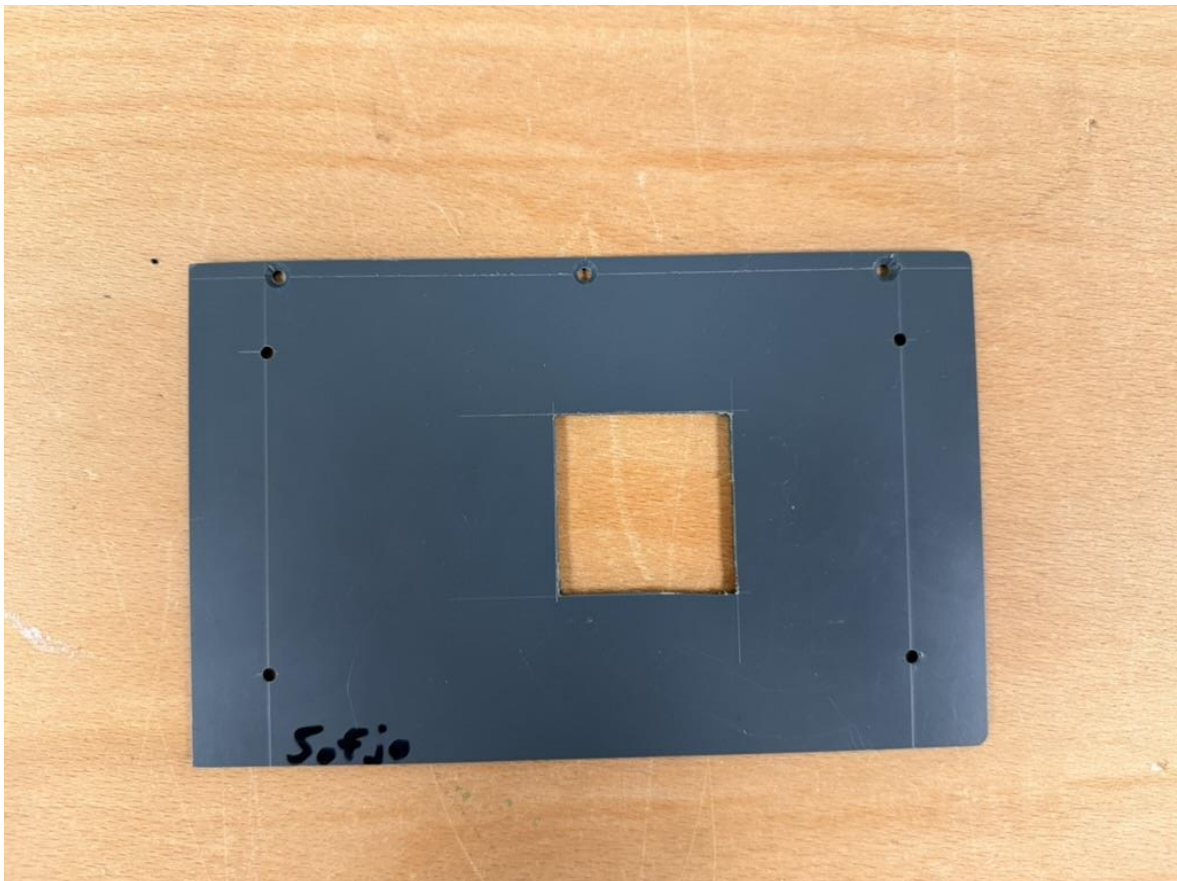
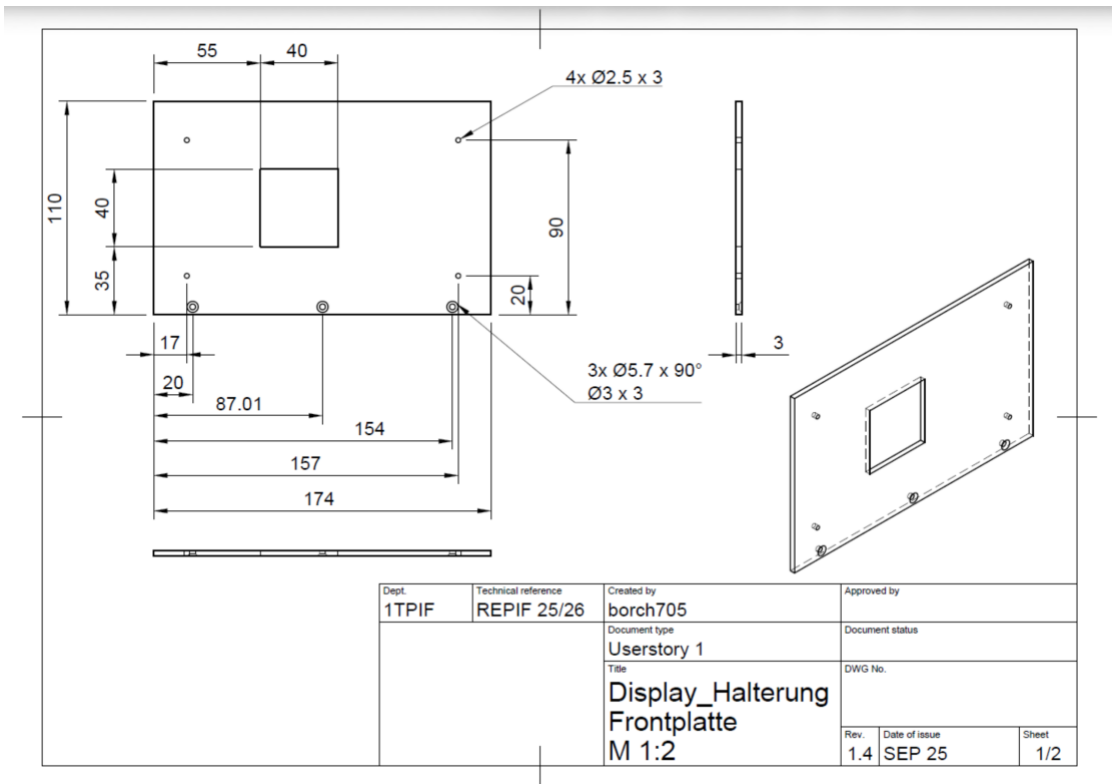
5.1 Connecting Both PVC Plates

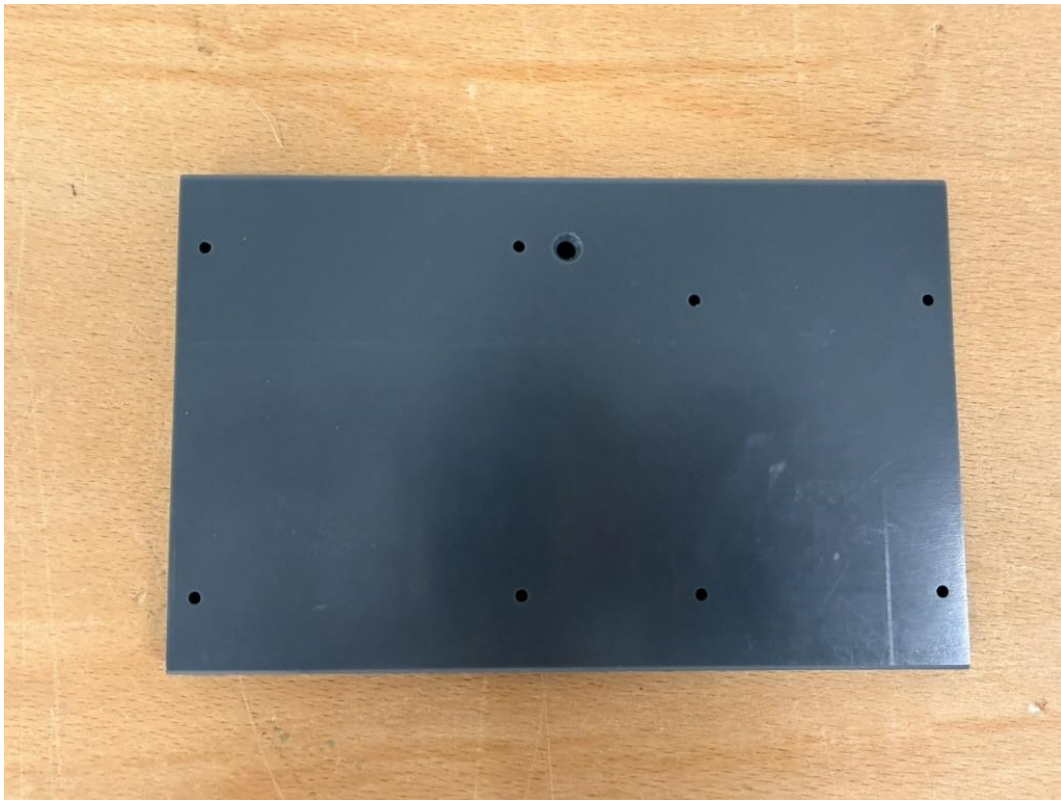
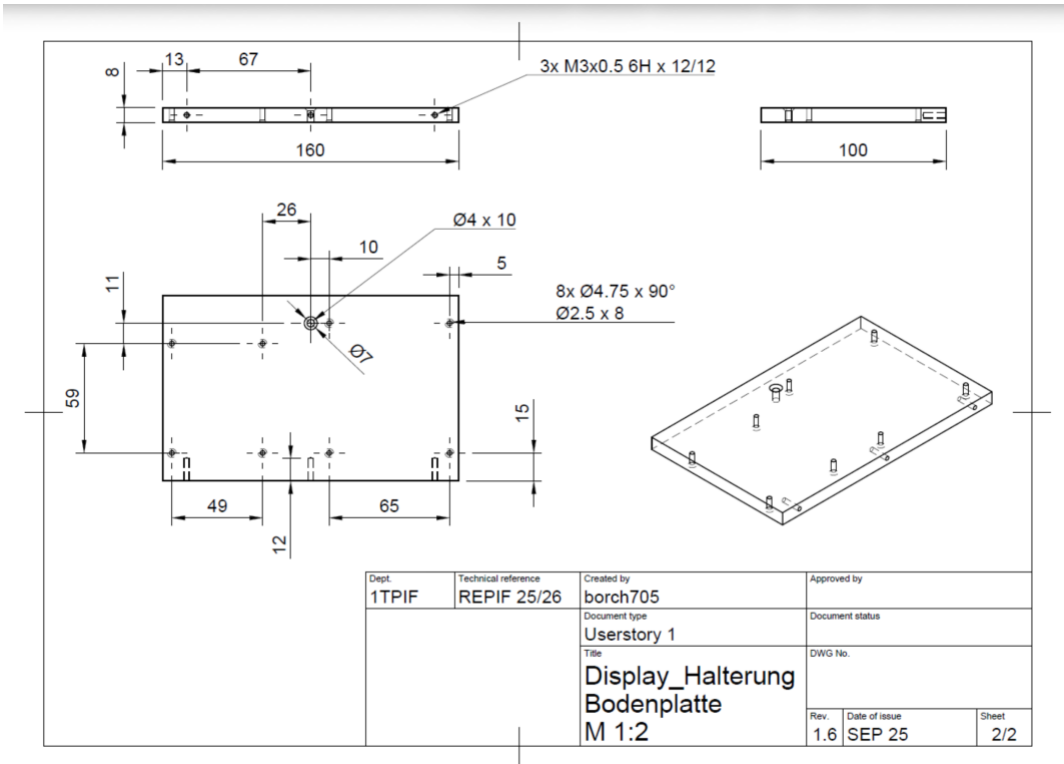
- The display holder and base plate were fastened together using **M3 countersunk screws**.

8. Result

The final product consists of:

- A precisely manufactured PVC display mount
- A drilled and threaded PVC base plate
- A correctly installed touch display and Raspberry Pi
- A fully assembled module ready to be installed into the wooden housing





(p.s. Next userstory will have more photoes of every step I will make)